Condenser Sizing App — User Manual (by Srk4on)

App link: https://condenser-app-jxcywsbg4ndwjc4qpitlnv.streamlit.app/

Github Link: https://github.com/Rohit-o/condenser-app (backend data)

0) Purpose of the tool

This app estimates:

- **Heat load (Q)** for condensation (and sensible cooling, as per implemented equation)
- **Distillation time**
- **Utility-side performance** for different utilities and pipe sizes
- **LMTD** (Log Mean Temperature Difference)
- **HTA (Heat Transfer Area)** in **m²**
- A **recommended (first-feasible)** Utility + Pipe diameter combination based on a **ΔT limit**
  This for people working in plant and want to have basic estimate about condenser sizing and adequacy.

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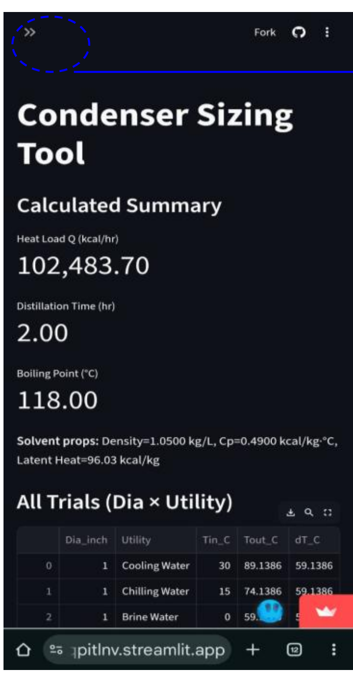
1) Quick orientation (Phone vs Laptop)

Phone view

What you will notice first: the input panel is hidden to save space.

- Tap the **<<** (double-arrow) icon near the top-left to **open/close the Inputs sidebar**.
- After opening the sidebar, you can select solvent and enter all process/design inputs.

1. **Insert Image (Phone – Inputs sidebar with << icon visible):**



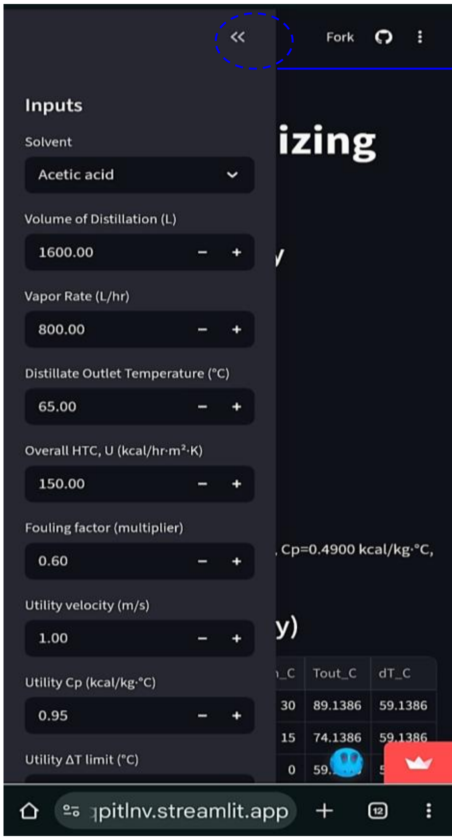
<< This icon is for inputs.

Where you can give input about solvent.

For giving input look next image

After giving input you can have a look to Distillation Time (hr).

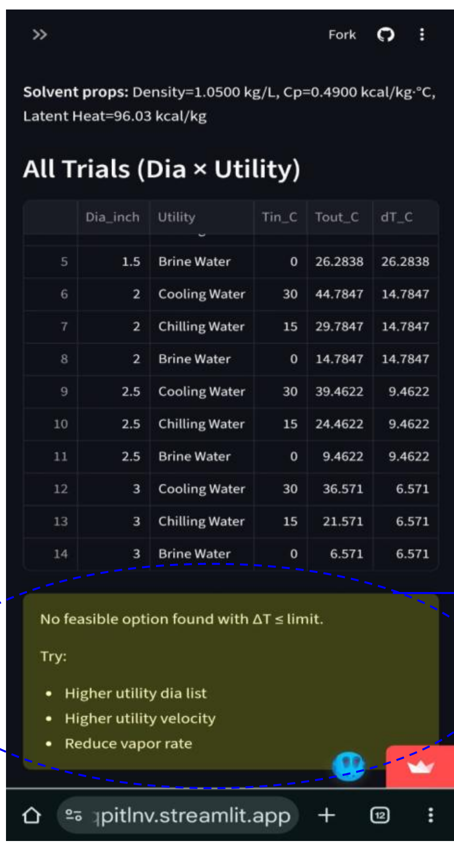
2. Insert Image (Phone – Summary + start of Trials):



<< This icon give idea about inputs. For Non-Engineers working in plant only four inputs are okay to manipulate.

1. Solvent
2. Volume of Distillate
3. Vapor Rate
4. Distillate Outlet Temperature

3. Insert Image (Phone – Trials + "No feasible option" message):

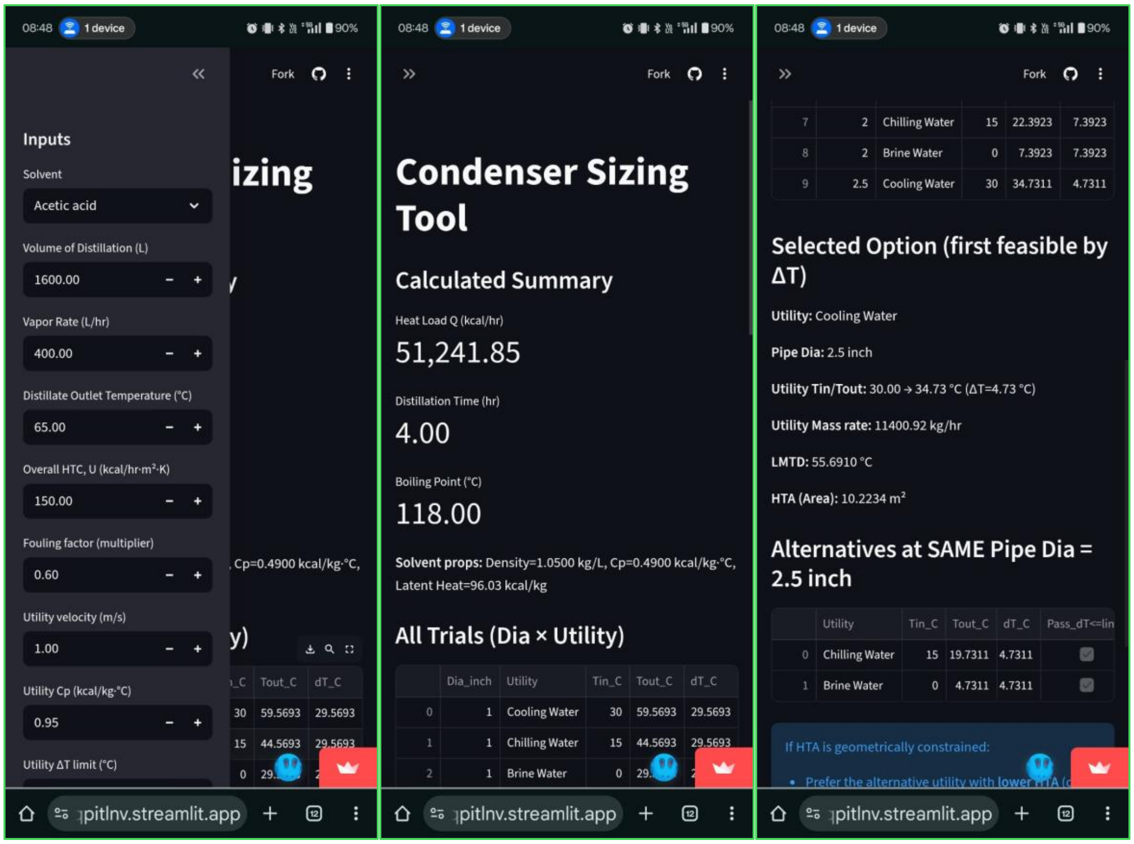


Once inputs are given. Now press enter. You can see the solution. It may happen you see no feasible solution.

So it gives you suggestion of **reducing vapor rate** plant person can reduce that again going back into the inputs.

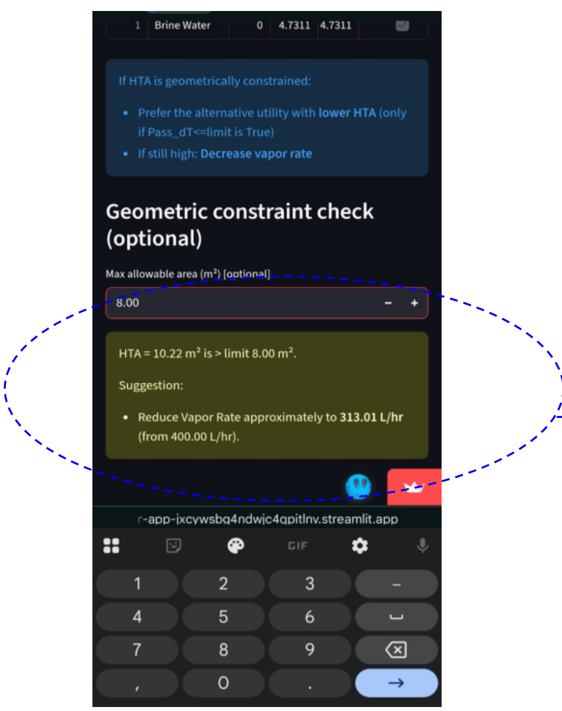
So let's decrease the vapor rate from 800 L/hr to 400 L/hr

4. Insert Image (Phone – Selected option + Alternatives):



Now we have the Utility Pipe diameter, YUUPP !! But wait, we might have heat exchanger area then, what vapor rate we should keep, with same utility dia that we calculated ??

5. Phone – Geometric constraint check:



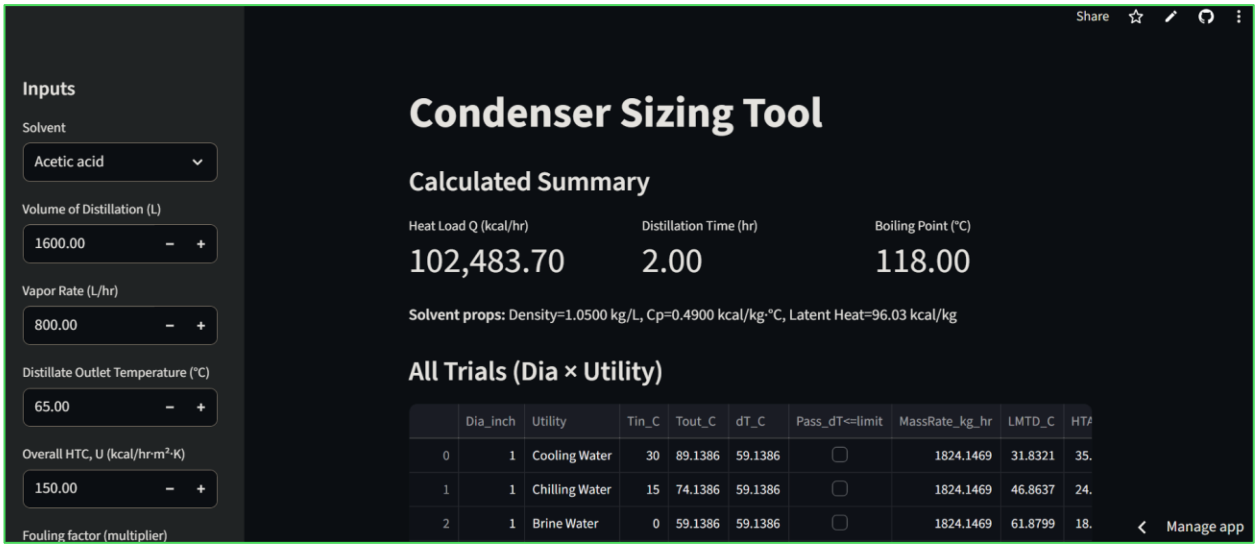
Now here we can see the answer was 10.22 m^2 but the user has 8 m^2 area Heat exchanger available with him/herself. So now the system has calculated adequate vapor rate for the calculated dia.

Laptop/Desktop view

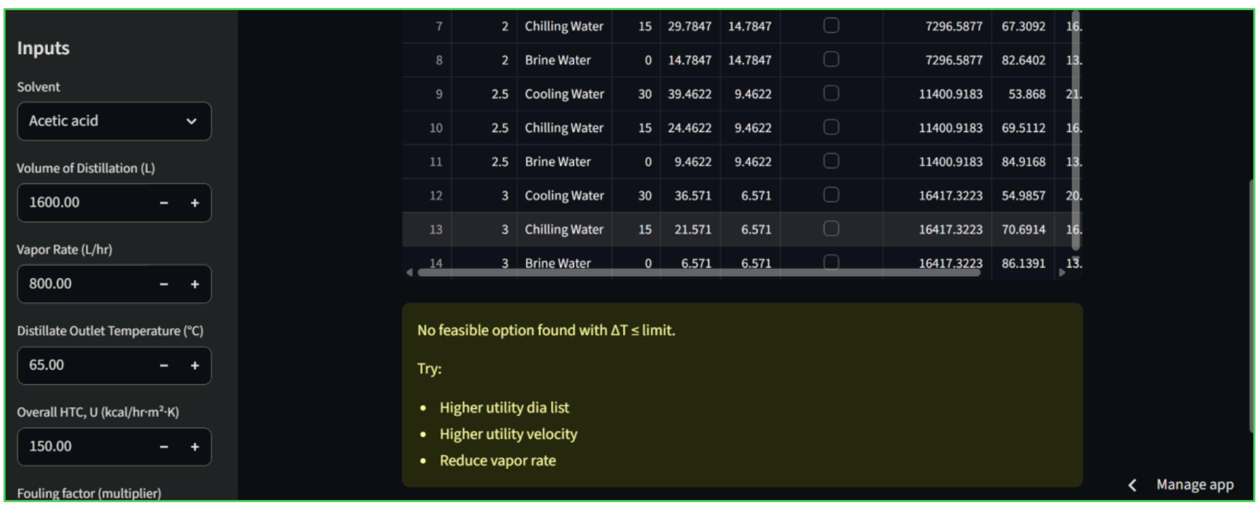
What you will notice first: the Inputs sidebar is visible on the left by default.

- Inputs are always accessible (no need to tap <<).
- Tables are easier to scroll and read on a wide screen.

1. **Laptop – Full page with Inputs + Summary + Trials:**



2. Insert Image (Laptop – Trials table + Warning message:



Now, again same story, **reduce the vapor rate.**

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2) Key assumptions & constraints (Important)

2.1 Heat exchanger assumption

- **Counter-current heat exchanger** is assumed for **LMTD** calculation.

2.2 Utility selection logic

- The app checks utilities in this order: **Cooling Water** → **Chilling Water** → **Brine Water**.

2.3 Pipe diameter constraint

- Available utility pipe diameters are limited to:
  - **1, 1.5, 2, 2.5, 3 inch**
- **Maximum diameter available in the app is 3 inch.**

2.4 Feasibility criterion

- A trial is considered feasible if:
  - **Utility ΔT ≤ ΔT limit** (default 5°C), and
  - **Driving force is positive** (LMTD is valid)

2.5 Temperature driving-force rule

- **Distillate outlet temperature must be greater than the maximum utility inlet temperature**.
- If not, the app stops and asks you to increase the distillate outlet temperature.

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3) Inputs (Left sidebar)

3.1 Solvent

- Choose from the solvent dropdown.
- The app pulls solvent properties from `data/condenser_backend.json`.

Solvent properties used: - **Boiling Point (°C)** - **Latent Heat (kcal/kg)** - **Density (kg/L)** - **Cp (kcal/kg·°C)**

3.2 Process inputs

- **Volume of Distillation (L)**
- **Vapor Rate (L/hr)**
- **Distillate Outlet Temperature (°C)**

How they affect results (quick): - Higher **Vapor Rate** → higher **Q** → higher **HTA** - Higher **Distillate Outlet Temp** → changes sensible term and driving force (can reduce/increase HTA depending on case)

3.3 Design inputs

- **Overall HTC, U (kcal/hr·m²·K)**
- **Fouling factor (multiplier)**

Interpretation: - Higher **U** → lower required area - Lower **fouling factor** → higher required area (more conservative)

3.4 Utility hydraulics inputs

- **Utility velocity (m/s)**
- **Utility Cp (kcal/kg·°C)**
- **Utility ΔT limit (°C)**

Interpretation: - Higher **velocity** → higher utility mass flow → lower utility ΔT - Lower **ΔT limit** makes selection stricter

Insert Image (Velocity / Utility Cp / ΔT limit): *(paste here)*

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4) Outputs & how to read them

4.1 Calculated Summary

Shows: - **Heat Load Q (kcal/hr)** - **Distillation Time (hr)** - **Boiling Point (°C)** - Solvent properties summary line

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4.2 All Trials (Dia × Utility)

This table lists results for every combination tested:

Columns you will typically use: - **Dia_inch**: pipe diameter - **Utility**: Cooling / Chilling / Brine - **Tin_C, Tout_C, dT_C**: utility inlet, outlet, and rise - **Pass_dT<=limit**: whether utility ΔT is within limit - **MassRate_kg_hr**: utility mass flow - **LMTD_C**: driving force for heat transfer - **HTA_m2**: required heat transfer area

How to interpret quickly: - Start from smallest diameter. - Look for the **first row** where **Pass_dT<=limit** is True and LMTD/HTA are valid.

4.3 “No feasible option found…” message

If all options fail (even at 3 inch), you will see a warning with suggestions:

- Increase diameter range beyond 3 inch (if possible) *{Future Work}*
- Increase utility velocity
- Reduce vapor rate

4.4 Selected Option (first feasible by ΔT)

This section shows the recommended selection: - Utility - Pipe diameter - Utility Tin → Tout and ΔT - Utility mass rate - LMTD - HTA (Area)

4.5 Alternatives at SAME Pipe Diameter

This section appears when the **Selected Utility is Cooling Water**.

Purpose: - If the calculated **HTA is geometrically large**, you can check **Chilling Water** and **Brine Water** at the **same selected diameter**. - Colder utilities often reduce required HTA (area).

4.6 Geometric constraint check (optional)

If plant layout/space limits the condenser area:

- Enter **Max allowable area (m^2)**
- If calculated HTA exceeds this limit, the app suggests a reduced vapor rate.

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5) Recommended workflow for users

1. Select solvent
2. Enter process inputs (Volume, Vapor Rate, Distillate outlet temperature)
3. Keep U and fouling factor as default unless you have plant-specific values
4. Review **All Trials**
5. Use **Selected Option** as baseline
6. If HTA is too large, use **Alternatives at same diameter** (if shown)
7. If still large, use **Geometric constraint check** and reduce vapor rate

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6) Troubleshooting

Issue A: “Distillate outlet temperature must be greater than max utility inlet”

- Increase distillate outlet temperature above the highest utility inlet temperature.
- Here Room Temperature is 30ºC, thus distillate outlet temperature must be greater than 30.

Issue B: “No feasible option found with ΔT ≤ limit”

Try (in order): - Increase utility velocity - Reduce vapor rate - Expand pipe diameter list beyond 3 inch (future enhancement)

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7) Change log (optional)

- v1: First public user manual draft (includes phone and laptop orientation + all app sections)

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8) Future work

- v2: It has limit on utility pipe diameter. In future I will extend this.
- v2: I will info about heat transfer coefficient based on the SS MS GL Reactor type and fluid flowing in shell (such as organic vapor) and tube (such as: water or oil)
- v2: I will incorporate pass effect. As this model only considered energy based calculation, to get the tentative idea of the people working in plant.